

Over the past 75 years the study of kaon physics has been integral to the development of the standard model. Since 2016, the NA62 experiment has been the leading experiment in the field of kaon physics. However, the future of kaon physics is uncertain, with the NA62 experiment funded up to CERN LS3, which is expected at the end of 2025. It was then required to investigate the future of kaon physics and the potential experiments that could succeed NA62. This chapter will discuss the potential future of kaon physics, the experiments that could be used to further the field, and the work contributed to these experiments.

0.1 High Intensity Kaon Experiments (HIKE)

The High-intensity kaon experiments (HIKE) was a proposed project to continue kaon physics at CERN in the ECN3 experimental hall and represented a long-phased plan after LS3. HIKE would have profited from a four to six-fold increase in beam intensity compared to NA62 and included cutting-edge detector technologies that will be described in this section. There were three phases within the HIKE program with the first being a continuation of NA62's charged kaon physics with the high intensity beam and detector upgrades. The second would have involved neutral kaon physics with charged secondary particle tracking and the third phase would have been designed to measure the $K_L \rightarrow \pi^0 \nu \bar{\nu}$ decay.

0.1.1 Phase 1

0.1.2 Phase 2

0.1.3 Phase 3

0.1.4 RICH upgrade design

0.1.5 GTk upgrade design

0.1.6 Detector Rates